

AKSHAT GUPTA

CS PhD, UC Berkeley

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Berkeley, California  [Google Scholar](#)

SUMMARY

I'm a third-year PhD Student at UC Berkeley affiliated with BAIR. I'm currently working on continual learning and algorithmic discovery.

SKILLS

Languages: Python, Go, MATLAB, C++, Javascript, C

Tools: Pytorch, Huggingface, NLTK, Spacy

Technologies: AWS, GCP, Azure, Dialogflow, Docker

EDUCATION

8/2023 - Present	PhD, Computer Science Advisor : Gopala Anumanchipalli	University of California, Berkeley
1/2020 - 5/2021	MS, Electrical and Computer Engineering Advisor : Alan W Black	Carnegie Mellon University
10/2017 - 3/2020	MS, Applied and Engineering Physics Thesis Advisor : Gregory Eyink (Johns Hopkins University)	Technical University of Munich, Germany
8/2013 - 6/2017	B.Tech, Electrical Engineering Thesis Advisor : Renu Rameshan	Indian Institute of Technology Mandi

EXPERIENCE

4/2025 - 8/2025	Student Researcher • Topic - Automated Discovery, Multi-Agent Systems, Code and Systems Optimization • Working Team - Chun-Liang Li (Manager), Rui Meng, Violet Peng	Google
5/2025 - 8/2025	Research Scientist Intern • Topic - Latent Reasoning in LLMs • Working Team - Payel Das (Manager), Subhajt Chaudhury, Matt Riemer	IBM
6/2021 - 8/2023	Senior Associate, AI Research • Topics - Relation extraction, Knowledge graphs and Large scale public information discovery • Publication - [16], Patents - [1], [2], [3],	JPMorgan Chase
9/2018 - 6/2019	Visiting Research Scholar • Topic - Vorticity dynamics in turbulent channel flows operating under Lagrangian Dynmaics • Publications - [20, 21]	Johns Hopkins University

PUBLICATIONS

- Automated Discovery has No Universally Superior Harness**
Akshat Gupta, Alexander Lu, Jermaine Lei, Gopala Anumanchipalli, Leshem Chosehn
Under Review
- AdaEvolve: Adaptive LLM Driven Zeroth-Order Optimization**
Mert Cemri*, Shubham Agrawal*, Akshat Gupta, Shu Liu, Audrey Cheng, Qiuyang Mang, Ashwin Naren, Lutfi Eren Erdogan, Koushik Sen, Matei Zaharia, Alex Dimakis, Ion Stoica
Under Review
- Evolutionary Strategies lead to Catastrophic Forgetting in LLMs**
Immanuel Abdi*, Akshat Gupta*, Micah Mok, Alexander Lu, Nicholas Lee, Gopala Anumanchipalli
ACL 2026 Main Conference Oral
- OrthoEdit: Principled and Stable Knowledge Editing via Orthogonal Subspace Projection**
Shanbao Qiao, Xuebing Liu, Akshat Gupta, Seung-Hoon Na
TACL
- Weight Tying Biases Token Embeddings Towards the Output Space**
Antonio Lopardo, Avyukth Harish, Catherine Arnett, Akshat Gupta
ACL 2026 Findings
- How do LLMs use their Depth?**
Akshat Gupta, Jay Yeung, Gopala Anumanchipalli, Anna Ivanova
Under Review
- Lifelong Knowledge Editing requires Better Regularization**
Akshat Gupta, Phudish Prateepamornkul, Maochuan Lu, Ahmed Alaa, Thomas Hartvigsen, Gopala Anumanchipalli
EMNLP 2025 Findings

8. **Efficient Knowledge Editing via Minimal Precomputation**
Akshat Gupta, Maochuan Lu, Thomas Hartvigsen, Gopala Anumanchipalli
ACL 2025 Main Conference
9. **Sylber: Syllabic Embedding Representation of Speech from Raw Audio**
Cheol Jun Cho, Nicholas Lee, **Akshat Gupta**, Dhruv Agarwal, Ethan Chen, Alan Black, Gopala Anumanchipalli
ICLR 2025
10. **PokerBench : Training Large Language Models to become Professional Poker Players**
Richard Zhuang, **Akshat Gupta**, Richard Yang, Aniket Rahane, Zhengyu Li, Gopala Anumanchipalli
AAAI 2025
11. **Norm Growth and Stability Challenges in Localized Sequential Knowledge Editing**
Akshat Gupta, Christine Fang, Atahan Ozdemir, Maochuan Lu, Ahmed Alaa, Thomas Hartvigsen, Gopala Anumanchipalli
KnowFM @ AAAI 2025 (Outstanding Paper Award)
12. **Rebuilding ROME : Resolving Model Collapse during Sequential Model Editing**
Akshat Gupta, Sidharth Baskaran, Gopala Anumanchipalli
EMNLP 2024 Main Conference
13. **A Unified Framework for Model Editing**
Akshat Gupta, Dev Sajjani, Gopala Anumanchipalli
EMNLP 2024 Findings
14. **Model Editing at Scale leads to Gradual and Catastrophic Forgetting**
Akshat Gupta, Anurag Rao, Gopala Anumanchipalli
ACL 2024 Findings
15. **Self-Assessment Tests are Unreliable Measures of LLM Personality**
Akshat Gupta, Xiaoyang Song, Gopala Anumanchipalli
BlackboxNLP 2024, co-located with EMNLP 2024
16. **REFinD: Relation Extraction Financial Dataset**
Simerjeet Kaur, Charese Smiley, **Akshat Gupta**, Joy Sain, Dongsheng Wang, Toyin Aguda, Sameena Shah
SIGIR 2023
17. **Probing Quantifier Comprehension in Large Language Models: Another Example of Inverse Scaling**
Akshat Gupta
BlackboxNLP 2023, co-located with EMNLP 2023
18. **Intent classification using pre-trained language agnostic embeddings for low resource languages**
Hemant Yadav, **Akshat Gupta**, Sai Krishna Rallabandi, Alan W Black, Rajiv Ratn Shah
Interspeech 2022
19. **Acoustics Based Intent Recognition Using Discovered Phonetic Units for Low Resource Languages**
Akshat Gupta, Sai Krishna Rallabandi, Alan W Black
ICASSP 2021
20. **Stochastic Lagrangian dynamics of vorticity. Part 1. General theory for viscous, incompressible fluids**
Gregory Eyink, **Akshat Gupta**, Tamer Zaki
Journal of Fluid Mechanics, 2020 (Volume 901, Page A2)
21. **Stochastic Lagrangian dynamics of vorticity. Part 2. Application to near-wall channel-flow turbulence**
Gregory Eyink, **Akshat Gupta**, Tamer Zaki
Journal of Fluid Mechanics, 2020 (Volume 901, Page A3)

PATENTS

1. **Method and system for automated public information discovery**
Akshat Gupta, Simerjot Kaur, Xiaomo Liu, Armineh Nourbakhsh, Andrea Stefanucci, Alex Woodgate, Sameena Shah
US Patent 12,001,491
2. **Method and system for detection of anomalous rejections of foreign exchange requests**
Nacho Navarro, Xiaomo Liu, Simran Lamba, **Akshat Gupta**, Sameena Shah
US Patent App. 18/085,104
3. **Method and system for understanding financial documents**
Simerjot Kaur, Charese Smiley, Joy Sain, Suchetha Siddagangappa, **Akshat Gupta**, Sameena Shah
US Patent App. 17/647,356

AWARDS

- Outstanding Paper Award, KnowFM Workshop @ AAAI 2025 [\[11\]](#)

- NVIDIA Compute Grant of 16,000 GPU hours for A100 GPUs (80GB) on the topic - "Enhancing Post-training Editability via Knowledge Modularization"
- Student Travel Grant, BlackboxNLP 2024, co-located with EMNLP 2024
- UC Berkeley EECS Fellowship, 2023
- Best Undergraduate Thesis, IIT Mandi 2017 [Topic - Blind Image Deconvolution]

INVITED TALKS

- UC Berkeley, 4/2026 - *"Continual Learning : Learning during Problem Solving"* (Guest Lecture for CS 288 - Advanced Natural Language Processing)
- Cognizant AI Lab, 2/2026 - *"Evolutionary Strategies lead to Catastrophic Forgetting in LLMs"*
- Comet's Live Paper Reading, 1/2026 - *"POKERBENCH: Training Large Language Models to become Professional Poker Players"* [\[Link\]](#)
- UC Berkeley, 5/2025 - *"AI Interpretability"* (Guest Lecture for CS 188 - Introduction to Artificial Intelligence)
- A5 Labs, 4/2025 - *"LLMs and Poker"*
- UPitt NLP Seminar, 3/2025 - *"The Past, Present and Future of Knowledge Editing"*
- CMU, 11/2021 - *"Transformers and GNNs"* (Guest Lecture for 11785 - Introduction to Deep Learning) [\[Link\]](#)

ACADEMIC SERVICE

Reviewing

- Nature Machine Intelligence
- ICLR 2026, ICLR 2025, ICML 2025, COLM 2025
- ACL Rolling Review : June 2024, Aug 2024, Oct 2024, Dec 2024, May 2025, Oct 2025
- ACL 2023
- EMNLP 2022

Workshop Organization

- 4th Workshop on Knowledge Discovery from Unstructured Data in Financial Services, SIGIR 2023

ADVISING

- | | |
|--|----------------|
| • Jermaine Lei, Undergrad UC Berkeley, Published [1] | 2026 - Present |
| • Alexander Lu, Undergrad UC Berkeley, Published [3, 1] | 2025 - Present |
| • Maochuan Lu, Undergrad UC Berkeley (-> MS, CMU), Published [8, 11] | 2024 - 2025 |
| • Richard Zhuang, Undergrad UC Berkeley (-> MS, Stanford University), Published [10] | 2024 - 2025 |
| • Nehal Sindhu, Undergrad UC Berkeley (-> SWE, Amazon) | 2024 - 2025 |
| • Atahan Ozdemir, Undergrad UC Berkeley (-> SWE, Google), Published [11] | 2024 |
| • Dev Sajnani, Undergrad UC Berkeley, Published [13] | 2024 |
| • Anurag Rao, Undergrad UC Berkeley (-> MS, University of Oxford), Published [14] | 2023 - 2024 |
| • Xiaoyang Song, MS Columbia University (-> PhD, University of Michigan), Published [15] | 2023 |
| • Anant Singh, MS NYU (-> MLE, Apple) | 2022 - 2023 |